

# Jiaqi Wu 吳佳齊

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## RESEARCH INTERESTS

Computational and comparative genomics, molecular evolution, statistical genetics, and bioinformatics. Methods, benchmarks, and open-source software that make large-scale genomic analyses comparable and reproducible (SplitAligner, Catnip10, SGV-Caller); applications to animal genetics and breeding, genomic monitoring of wild mammals, mammalian life-history evolution, and viral genome surveillance.

## EDUCATION

- 2015.04 – 2018.03 **Ph.D. (Agriculture)**, Statistical Genetics, Graduate School of Agricultural and Life Sciences, The University of Tokyo (advisor: Prof. Hirohisa Kishino). Thesis: *Statistical modeling of species and population level genomic evolution*. Research student, 2014.10 – 2015.03.
- 2010.09 – 2013.06 **M.Sc.**, Bioinformatics, School of Life Sciences, Fudan University, China (advisors: the late Prof. Yang Zhong and Prof. Masami Hasegawa)
- 2004.09 – 2009.06 **B.Med.**, Preventive Medicine, Qilu School of Medicine, Shandong University, China

## POSITIONS

- 2025.04 – present Assistant Professor, School of Applied Biological Science / Graduate School of Integrated Sciences for Life, Hiroshima University
- 2024.04 – 2025.03 Researcher (CAP), School of Applied Biological Science, Hiroshima University
- 2021.04 – 2023.03 Project Researcher (JST CREST), Tokai University School of Medicine
- 2018.09 – 2021.03 JSPS Research Fellow (PD), School of Life Science and Technology, Tokyo Institute of Technology
- 2018.04 – 2018.08 Researcher, Tokyo Institute of Technology; Visiting Researcher, Research Institute of Evolutionary Biology, Tokyo
- 2009.07 – 2010.08 Laboratory Technician, Institutes of Biomedical Sciences, Fudan University & Shanghai Public Health Clinical Center

## SOFTWARE

- **SplitAligner** — branch-identity coordinate system for phylogenomics under missing taxa and gene-tree discordance (Perl; R track **SplitAlignerR** in development). [github.com/wujiaqi06/SplitAligner](https://github.com/wujiaqi06/SplitAligner)
- **Catnip10** — deterministic 10-tip conformance benchmark with an independent graph-contraction oracle (Python/R, MIT). [github.com/wujiaqi06/Catnip10](https://github.com/wujiaqi06/Catnip10)
- **SGV-Caller** — SARS-CoV-2 genome-variation caller used as a data source for the G2P-Japan consortium (Perl, MIT). [github.com/wujiaqi06/SGV-caller](https://github.com/wujiaqi06/SGV-caller)

## GRANTS

- 2022 – 2024 JSPS KAKENHI Grant-in-Aid for Scientific Research (B) 23K23950 and (C) 22K11950 — co-investigator
- 2018 – 2020 JSPS KAKENHI Grant-in-Aid for JSPS Fellows — principal investigator
- 2015 – 2016 Sasakawa Scientific Research Grant, The Japan Science Society — principal investigator
- 2012 – 2013 Fudan University Graduate Innovation Fund — principal investigator

## AWARDS

- 2018 Dean's Award, Graduate School of Agricultural and Life Sciences, The University of Tokyo
- 2012 National Scholarship, Ministry of Education of the People's Republic of China; Lu Zonglin Scholarship and First-class Academic Scholarship, Fudan University

## TEACHING & SUPERVISION

- Hiroshima University (2025 –): Animal Genetics and Breeding; Animal Production Science Laboratory I; International Applied Biological Sciences I; Introductory Physiology of Domestic Animals (incl. AIMS programme).
- Guest lectures: The University of Tokyo, Agricultural Bioinformatics Research Unit, “Statistics of Biological Sequences” (2019); Kimura Motoo Memorial Evolution Seminar (2018); Fudan University summer school (molecular evolution and phylogenetics).
- Co-supervisor of doctoral students at Hiroshima University; research supervision of master's students (Fudan University) leading to co-authored papers.

## PROFESSIONAL SERVICE & SOCIETIES

Member of The Genotype to Phenotype Japan (G2P-Japan) consortium. Society of Evolutionary Studies, Japan; Ecological Society of Japan; Japanese Society of Animal Breeding and Genetics; Society for Molecular Biology and Evolution (SMBE); Society of Systematic Biologists (SSB); American Society of Human Genetics (ASHG).

**SELECTED PUBLICATIONS** (first / co-first / corresponding author; † co-first, \* corresponding. 40 peer-reviewed papers in total; Google Scholar citations 7,267, h-index 22, Aug 2026)

1. **Wu J**, Endo H, Rakotondraparany F, Ratsoavina FM, Imai K, Yamamoto Y, Shimogiri T, Mannen H, Yonezawa T. Re-evaluating the population history of Malagasy native cattle reveals shared African taurine-related ancestry but distinct indicine admixture histories relative to East African cattle. *Animal Science Journal* 97(1):e70232 (2026). doi:10.1111/asj.70232
2. Yonezawa T†, **Wu J**†, Masuko R, et al., Lenstra JA, Mannen H. Genome-wide variation reveal that goats were introduced into Asia via multiple migrations. *Scientific Reports* 16:163 (2026).
3. **Wu J**, Kryukov K, Takeuchi JS, Nakagawa S. SGV-caller: SARS-CoV-2 genome variation caller. *Heliyon* 11(4):e42613 (2025).
4. Wang T†, **Wu J**†\*, Nakagawa S, Yonezawa T, Liu Z, Zhang X, Wang H, Li Y, Zhang T\*. Evolutionary analysis and population dynamics in the global transmission of Kaposi's sarcoma-associated herpesvirus. *Archives of Virology* 170(5):92 (2025).
5. Kimura I†, Yamasoba D†, Nasser H†, Ito H†, Zahradnik J†, **Wu J**†, Fujita S†, Uriu K†, et al., Sato K. Multiple mutations of SARS-CoV-2 Omicron BA.2 orchestrate its virological characteristics. *Journal of Virology* 97(10):e0101123 (2023).
6. **Wu J**\*, Yonezawa T, Kishino H\*. Molecular evolutionary rate predicts intraspecific genetic polymorphism and species-specific selection. *Genes* 13(4):708 (2022).
7. Zhao X†, **Wu J**†, Kishino H, Chen L. Massive loss of transcription factors promotes the initial diversification of placental mammals. *International Journal of Molecular Sciences* 23(17):9720 (2022).
8. Kimura I†, Kosugi Y†, **Wu J**†, Zahradnik J†, et al., Sato K. The SARS-CoV-2 Lambda variant exhibits enhanced infectivity and immune resistance. *Cell Reports* 38:110218 (2022).
9. **Wu J**\*, Yonezawa T, Kishino H\*. Evolution of reproductive life history in mammals and the associated change of functional constraints. *Genes* 12(5):740 (2021).
0. Liu Z†, Mao X†, **Wu J**†, Yu K, et al., Chen X. World-wide prevalence of substitutions in HCV genome associated with resistance to direct-acting antivirals. *Clinical Gastroenterology and Hepatology* 19:1906 (2021).
1. **Wu J**\*, Yonezawa T, Kishino H\*. Rates of molecular evolution suggest natural history of life-history traits and a post-K–Pg nocturnal bottleneck of placentals. *Current Biology* 27:3025 (2017).
2. Cheng T†, **Wu J**†, Wu Y†, et al., Feng Q, Xia Q, Mita K. Genomic adaptation to polyphagy and insecticides in a major East Asian noctuid pest. *Nature Ecology & Evolution* 1:1747 (2017).
3. **Wu J**, Kohno N, Mano S, Fukumoto Y, Tanabe H, Hasegawa M, Yonezawa T. Phylogeographic and demographic analysis of the Asian black bear (*Ursus thibetanus*) based on mitochondrial DNA. *PLoS ONE* 10:e0136398 (2015).
4. **Wu J**, Hasegawa M, Zhong Y, Yonezawa T. Importance of synonymous substitutions under dense taxon sampling and appropriate modeling in reconstructing the mitogenomic tree of Eutheria. *Genes & Genetic Systems* 89:237 (2014).

Preprints (single author): **Wu J**. SplitAligner: a gene–species tree reconciliation framework using split-based branch mapping. *bioRxiv* (2026), doi:10.64898/2026.02.24.707838. · **Wu J**. A unique graph-theoretic truth table for cross-gene branch identity. *bioRxiv* (2026), doi:10.64898/2026.07.22.740066.

Full publication list: [wujiaqi06.github.io/#publications](https://wujiaqi06.github.io/#publications) · Last updated August 2026.